Lab 5: Proteins

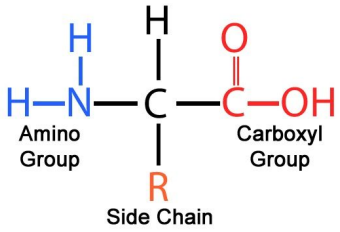
Agenda

- Mini-Lecture
  - Proteins
  - Cross-Contamination and Cleaning
- Recipes and Group Assignments
- Cooking Time (until 7pm)
- Discussion (7:00-7:10)
- Clean up (7:10-7:30)

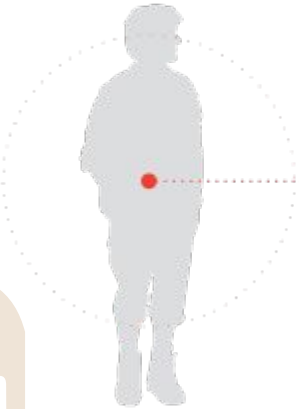
Mini Lecture

Proteins

What is protein?



- ★ Proteins are groups of amino acids bonded and folded together.
- ★ Protein supports muscle growth, immune function, DNA regulation, hormonal processes, digestion, and the distribution of essential molecules such as oxygen throughout the body.



NON ESSENTIAL	ESSENTIAL
Alanine	Histidine
Arginine	Isoleucine
Asparagine	Leucine
Aspartate	Lysine
Cystine	Methionine
Glutamic	Phenylalanine
Glycine	Threonine
Ornithine	Tryptophan
Proline	Valine
Serine	
Tyrosine	

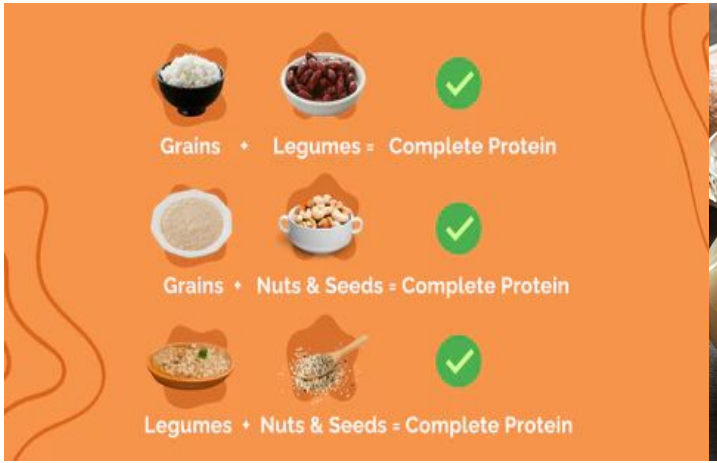
Think Quantity AND Quality

Incomplete Proteins

- Missing one or more of the 9 essential amino acids
- Sources: legumes, nuts, seeds, whole grains

Complete Proteins

- Have all 9 essential amino acids
- Sources: animal products, soy, quinoa, buckwheat, hemp seeds, nutritional yeast



Think Quantity AND Quality

- ★ 11 grams protein per 100 calories broccoli
- ★ 2 grams protein per 1 cup broccoli

- ★ 9 grams protein per 100 calories steak
- ★ 42 grams protein per 1 cup steak

Mini Lecture

Cooking Proteins Safely

Preventing Cross-Contamination

★ **Cross-contamination:** the transfer of harmful bacteria between food products

Remember our 4 steps of food safety?

1. **Clean -** Start with clean hands, utensils and surfaces. Wash your hands every time you switch tasks (e.g. moving from cutting meat to cutting vegetables). If wearing gloves, simply taking them off does not count.

2. **Separate -** Keep raw animal products separate from ready-to-eat foods. Use different cutting boards, knives, utensils, etc.

3. **Cook -** Cook your food to its safe minimum internal temperature, which you should measure with a food thermometer.

4. **Chill -** Keep perishable foods at a safe temperature (below 40°F). Follow the two-hour rule and do not leave perishable foods at room temperature for more than two hours (one hour if the food is sitting in temperatures above 90°F). Plan to use or freeze leftovers within four days of when you first cooked them. When you do eat leftovers, reheat them to 165°F.

Minimum Internal Temperatures

Product	Minimum Internal Temperature & Rest Time
Beef, Pork, Veal & Lamb Steaks, chops, roasts	145 °F (62.8 °C) and allow to rest for at least 3 minutes
Ground Meats	160 °F (71.1 °C)
Ground Poultry	165 °F
Ham, fresh or smoked (uncooked)	145 °F (62.8 °C) and allow to rest for at least 3 minutes
Fully Cooked Ham (to reheat)	Reheat cooked hams packaged in USDA-inspected plants to 140 °F (60 °C) and all others to 165 °F (73.9 °C).
All Poultry (breasts, whole bird, legs, thighs, wings, ground poultry, giblets, and stuffing)	165 °F (73.9 °C)
Eggs	160 °F (71.1 °C)
Fish & Shellfish	145 °F (62.8 °C)
Leftovers	165 °F (73.9 °C)
Casseroles	165 °F (73.9 °C)

Housekeeping

- Stainless pans can be cleaned with a sponge, brush, or **steel wool** for stubborn food.
- Metal tools should never be used on non-stick pans, as they will scrape off the non-stick coating.
- Only put food waste in the compost bin.
- Avoid putting food in the sinks. If food falls in, clean it out.
- Only grab what you need to minimize food waste.
- As usual, complete the cleaning steps listed on the whiteboard.

Cooking Time

Group Assignments

Recipe Notes

- **Everyone with two recipes** - Divide and conquer! Work on both recipes at the same time.
- **Pesto Chicken & Zoodles** - I will visit to your station to demonstrate using the mandoline and food processor.
- **Turkey and Bulgur Patties** - We are using quinoa instead of bulgur.
- **Veggie Sliders** - Only make 12 patties (the number the recipe says it yields), and please cut your cheese slices in half or fourths.
- **Mini Quiche / Tofu Adobo** - We are not making sinangag (garlic fried rice) today, so please ignore that recipe.

Discussion

1. How did different cooking methods (sautéing vs baking) affect the texture and flavor of the food?
2. Did you find any challenges with cooking the proteins (overcooking, undercooking, moisture, etc.)? How did you address them?
3. Were you aware of incomplete vs complete proteins before this class? Can anyone tell me 2 ways plant foods can be consumed as complete proteins?
4. What protein dish did you enjoy most, and why? Was it due to the texture, flavor, or cooking method of the dish?

Next Lab:

Lab 6: Microwave Cooking

To Do:

1. Bring your laptop to lecture on Monday for the in-class assignment

2. Quiz #3 due Monday at 11:59pm

- **Please arrive on time. We will be starting promptly at 10 after the hour.** Late arrival will only affect your cooking time.
- You are welcome to look over past lab recipes for inspiration beforehand. They are all posted in the Files section of BCourses, under "Lab Materials".
- The lab will be completed in **pairs of two**. You and your partner must come up with a meal prioritizing your assigned ingredient, and create a balanced plate including protein, carbohydrates, and vegetables.

  - Assigned ingredients will be face down at each station upon arrival. **Do not turn over until we start**.
  - You will have the first 5 minutes to plan your meal after you see your assigned ingredient (includes researching best cooking times for your planned meal– meats in the oven, can see the ingredients available to you, but do not grab until 60 minute starter, etc.)
  - You will then each have an hour to make your food. (I will pace you all with time checks).
  - Plate your meal just for you.
  - After the hour, there will be 10-15 minutes for open community/class discussion sharing responses to significant questions on the board.
  - Lastly, you and your partner will be given a station scorecard to complete. **To check out of the lab, you must properly clean your stations, and hand in your scorecard to me. If you cooking early, you can complete the scorecards ahead of time.**